Sufficient conditions for the non-separating paths

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Abstract

For two vertices in a graph, a path between them is non-separating if the remaining graph is connected. A previous work shows that a non-separating path always exists if the connectivity is more than two. In this paper, we show that to determine if there is a non-separating path is NP-complete for graphs of connectivity one or two. Sufficient conditions for the existence of a non-separating path on biconnected graphs are given, as well as polynomial time algorithms for finding such a path whenever the condition holds.

1 Introduction

Consider two disjoint nodes s and t in a network G. If we want to allocate a vertex subset for the dedicated communication between s and t, it is natural to choose an st-path. However, if the chosen nodes cannot be used for other services at the period of time, the remaining network may be no more connected. Let $G = (V, E)$ be a connected simple undirected graph. For two given vertices s and t, a st-path P is non-separating if $G-P$ is connected. To keep the remaining network connected, one may hope to find a non-separating path. However, in general, non-separating paths do not always exist.

The non-separating path problem has been studied from the perspective of graph theory. Most of the works are devoted to its relationship to graph connectivity [1, 2, 5, 6]. Particularly, in [2], it is shown that for a positive integer k, there is a minimum integer $\alpha(k)$ so that if G is an $\alpha(k)$-connected graph, then there exist k internal disjoint non-separating st-paths for arbitrary s and t. Furthermore,due to the works in [2, 8], it is known that $\alpha(1) = \alpha(2) = 3$, $\alpha(3) = 6$ and $\alpha(k) \leq 22k + 2$ for any $k > 3$. Due to [8], for any s and t in a 3-connected graph, a non-separating st-path always exists. But it is still interesting to know how to determine a non-separating path for graph of low connectivity.

In this paper, we show the following:

- The problem of determining if there exists a non-separating path is NP-complete.
- We present sufficient conditions that a non-separating path exists, as well as a polynomial time algorithm to find it.

The paper is organized as follows: In the next section, we give some definitions and notations used in this papas. In Section 3, we discuss the complexities on graphs of connectivities one and two. Sufficient conditions are shown in Section 4.

2 Definitions and notations

For two sets U and V, $U - V = \{v|v \in U, v \notin V\}$. Let $G = (V, E)$, $v \in V$, $S \subset V$ and H an induced subgraph of G. The subgraph induced by S is denoted by $G[S]$. By $G - S$ we denote $G[V-S]$. Similarly $G-H = G[V-V(H)]$, $G-v = G[V - \{v\}]$ and $H \cup S = G[V(H) \cup S]$, in which $V(H)$ is the vertex set of H. By $H + v$ we denote the induced subgraph $G[V(H) \cup \{v\}]$. A vertex bipartition $(U, V - U)$ is a *connected bipartition* if both $G[U]$ and $G[V - U]$ are connected. For a connected partition $(U, V - U)$, a vertex v is movable if moving v to the other part remains a connected bipartition. A trivial observation is that $v \in U$ is movable iff v is not an articulation vertex in $G[U]$ and v has a neighbor in $V - U$, assuming $V - U$ is not empty.

A vertex subset X is a vertex cut set, or briefly a cut, if $G - X$ is disconnected. We shall assume that a cut is always minimal, i.e., any its proper subset is not a cut. When $|X| = k$, we say it is a k-cut. When X contains only one vertex x, we omit the set notation and say "x is a cut" or "x is an articulation vertex". Furthermore if s and t are

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in different components of $G-X$, we say X is a *st*-cut, and otherwise a non-*st*-cut. Note that X is a non-*st* cut if $s \in X$ or $t \in X$. The non-separating path problem is formally defined as follows.

> PROBLEM: Non-separating Path Problem (NSP)
> INSTANCE: A connected graph G and two vertices s and t.
> GOAL: Determine if there exists a *st*-path P such that $G-P$ is connected.

We shall abuse the problem name as a predicate, i.e., $\mathrm{NSP}(G, s, t)$ is true iff there is a non-separating *st*-path in G. A Hamiltonian *st*-path in a graph G is a path from s to t and visiting each vertex of G exactly once. We define a predicate $\mathrm{Hpath}(G, s, t)$ which is true iff such a path exists.

3 NP-completeness

3.1 Graphs with articulation vertices

Suppose that a vertex x is a cut of G. Let C_s and C_t be the components of $G-x$ which contain s and t, respectively. The following result can be easily obtained.

Lemma 1: Let $x \notin \{s, t\}$ be a non-*st* cut.

- In the case that there are more than two components in $G-x$: $\mathrm{NSP}(G, s, t)$ iff $\mathrm{NSP}(C_s - x, s, t)$.
- In the case that there are exactly two components in $G-x$: $\mathrm{NSP}(G, s, t)$ iff $\mathrm{NSP}(C_s - x, s, t)$ or $\mathrm{NSP}(C_s, s, t)$.

Next we consider that s or t is a cut. We show the result of s, and it is similar for t.

Lemma 2: If s is a cut and there are more than two components in $G-s$, there is no non-separating *st*-path in G.

Lemma 3: If s is a cut and there are exactly two components in $G-s$, then $\mathrm{NSP}(G, s, t)$ iff $\mathrm{Hpath}(C_t + s, s, t)$.

Now we turn to the case that there exists a vertex which is a *st*-cut.

Lemma 4: If x is an *st*-cut and there are more than three components in $G-x$, there is no non-separating *st*-path in G.

Proof: In this case, x must be on the path. If there are more than one components except for C_s and C_t, they cannot be connected after removing x. □

Lemma 5: If x is an *st*-cut and there are exactly three components in $G-x$, then $\mathrm{NSP}(G, s, t)$ iff $\mathrm{Hpath}(G-C, s, t)$, in which C is the component containing neither s nor t.

Lemma 6: If x is an *st*-cut and there are exactly two components in $G-x$, then $\mathrm{NSP}(G, s, t)$ iff $\mathrm{Hpath}(C_s + x, s, x) \wedge \mathrm{NSP}(C_t + x, x, t)$ or $\mathrm{Hpath}(C_t + x, t, x) \wedge \mathrm{NSP}(C_s + x, x, s)$.

The above results also implies that the Hamiltonian path problem can be reduced to NSP problem. Since Hamiltonian path problem is NP-complete [4], we obtain the following theorem.

Theorem 7: The NSP problem on graphs of connectivity one is NP-complete.

3.2 Biconnected graphs

Next we show that the NSP problem on biconnected graphs is NP-complete. The transformation is from the 3-SAT problem and similar to the one in [9].

Let C_i, $1 \le i \le m$ be the clauses of the SAT problem and x_i, $1 \le i \le n$, the variables. We construct a 2-connected graph $G = (V, E)$ by the following two steps.

At the first step, let $V = \{s, t\} \cup \{C_j | 1 \le j \le m\} \cup X$, in which $X = \{x_i, \bar{x}_i | 1 \le i \le n\}$. Let $E = E_1 \cup E_2 \cup E_3$, in which $E_1 = \{(s, x_1), (s, \bar{x}_1), (t, x_n), (t, \bar{x}_n\}$, $E_2 = \{(x_i, x_{i+1}), (x_i, \bar{x}_{i+1}), (\bar{x}_i, x_{i+1}), (\bar{x}_i, \bar{x}_{i+1}) | \forall 1 \le i < n\}$, and $E_3 = \{(x_i, C_j) | \forall x_i \in C_j\} \cup \{(\bar{x}_i, C_j) | \forall \bar{x}_i \in C_j\}$. Figure 1 illustrates the graph G.

At the second step, we construct a graph G' from G as follows:

- For each pair of x_i and $\bar{x}_i$, we add two vertices x_{i1} and x_{i2} and four edges (x_i, x_{i1}), (x_i, x_{i2}), $(\bar{x}_i, x_{i1})$, and $(\bar{x}_i, x_{i2})$.
- We subdivide each edge in E_3 by adding a 4-cycle as shown in Figure 2.

Claim 8: If there is a truth assignment satisfying all the clauses, there is an non-separating *st*-path in G'.

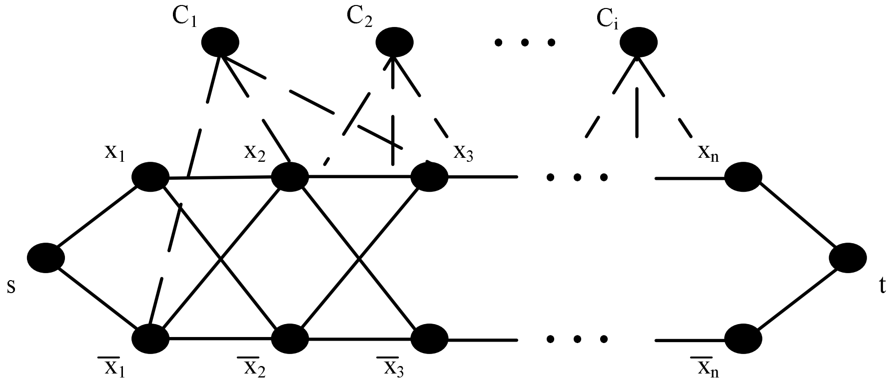

Figure 1: Transformation from 3-SAT to non-separating path (I)

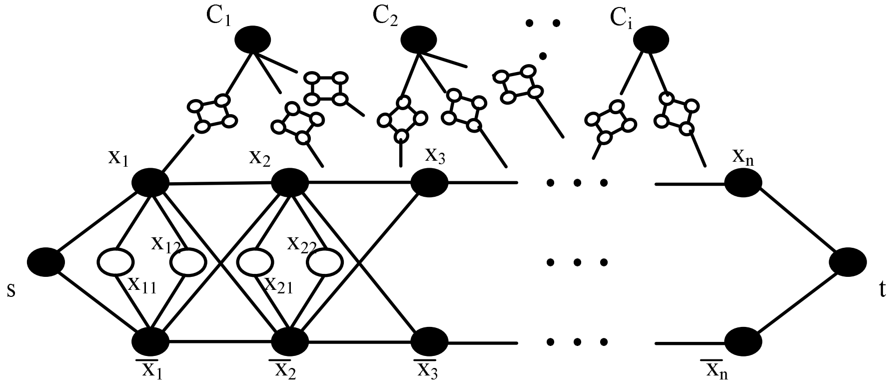

Figure 2: Transformation from 3-SAT to non-separating path (II)

Proof: For $1 \leq i \leq n$, let $v_i = x_i$ if x_i is assigned false, and $v_i = \bar{x}_i$ if x_i is assigned true. We claim $P = (s, v_1, v_2, \ldots, v_n, t)$ is a non-separating path as follows. Apparently P is a st-path passing through either x_i or $\bar{x}_i$ for $1 \leq i \leq n$. By the construction of G', the subgraph induced by $X - V(P)$ is connected. Since the truth assign satisfies all clauses, each C_j is connected to at least one vertex in $X - V(P)$, as well as all vertices added in the second step. □

Claim 9: If there is an non-separating st-path P on G', there is a truth assignment satisfying all the clauses.

Proof: For $1 \leq i \leq n$, the insertion of x_{i1} and x_{i2} st the second step makes any non-separating st-path cannot contain both x_i and $\bar{x}_i$. Similarly any non-separating st-path cannot pass through any C_j. Therefore, if P is a non-separating st-path on G, P contains, and only contains, either x_i or $\bar{x}_i$ for $1 \leq i \leq n$. Furthermore, since it is non-separating, all C_j are connected to $X - V(P)$.

For each $1 \leq i \leq n$, we assign x_i false if $x_i \in V(P)$ and assign true otherwise. Since each C_j are connected to $X - V(P)$, it is connected to some x_i or $\bar{x}_i$ which is assigned true. Consequentially, this truth assignment satisfies all the clauses. □

Theorem 10: The non-separating path problem on biconnected graphs is NP-complete.

Proof: Clearly the transformation can be done in polynomial time. It is also trivial G' is biconnected since each clause in a 3-SAT problem contains three literals.

By Claims 8 and 9, if there is a polynomial time algorithm for the non-separating path problem, it can also solve the 3-SAT problem. The result follows the NP-completeness of the 3-SAT problem [4]. □

4 Sufficient conditions for NSP

In this section we show a sufficient condition for the existence of a non-separating *st*-path. We first introduce an algorithm, and then the theorem is obtained by showing that the algorithm finds a non-separating path whenever the condition holds. Finally we give a more general sufficient condition.

Algorithm NSPA
Input: A biconnected graph G with no non-*st* 2-cut and two vertices s and t.
Output: A non-separating *st*-path.

choose an arbitrary vertex $v \notin \{s,t\}$;
$U \leftarrow \{v\}$;
repeat forever
    if there exists a movable vertex $u \in V - U - \{s,t\}$;
        move u from $V - U$ to U;
    else
        return $G[V - U]$.

Lemma 11 : Algorithm NSPA finds a non-separating *st*-path correctly in $O(|V||E|)$ time.

Proof: The algorithm always keeps two invariants:

- $(U, V - U)$ is a connected bipartition and
- $\{s,t\} \subset V - U$.

It repeatedly moves movable vertices other than s and t to U until no vertex can be moved without violating the two invariants. As shown in [3], for any connected bipartition of a biconnected graph, there are at least two movable vertices in each part. Therefore the algorithm terminates when s and t are the only movable vertices in $V - U$.

Let $H = G[V - U]$. A block of a graph is a maximally biconnected subgraph or an edge. Let $\mathcal{B}$ be the blocks of H and A be the cut vertices of H. The *block-cut tree* T of H is defined as follows. $V(T) = \mathcal{B} \cup A$, and for any $a \in A$ and $B \in \mathcal{B}$, $(a, B) \in E(T)$ iff a is a vertex of B in graph H. By definition T is a tree.

Since G is biconnected, if B is a leaf of T, there is a node in the block B of H which is movable with respective to the connected bipartition $(V - U, U)$. Since s and t are movable and there is no other movable vertex in H, T is a path. We shall show that each block B is an edge, and the proof is completed.

Suppose by contradiction that B is a block of H and $|V(B)| > 2$. Let D_i, $1 \le i \le r$, be the components of $H - B$. Since T is a path, any cut vertex of H belongs to at most two blocks, and therefore different D_i is adjacent to different vertex in B. Let v_i the vertex in B adjacent to D_i. Also due to that T is a path, we have $r \le 2$. We divided into the following cases. Note that any vertex in $V(B) - \{s,t\} - \{v_i | i \le r\}$ cannot be adjacent to $G - H$, or otherwise it will be movable since no vertex in a biconnected subgraph is a cut.

- $r = 0$: We have $\{s,t\} \subset V(B)$. Then $\{s,t\}$ is a 2-cut of G, a contradiction.
- $r = 1$: Either s or t is in B. W.o.l.g. we assume $s \in V(B)$. Then $\{s, v_1\}$ is a non-*st* 2-cut, a contradiction.
- $r = 2$: We have $\{s,t\} \cap V(B) = \emptyset$. Otherwise D_1 or D_2 can be moved into $G - H$. Then, similar to the case $r = 1$, $\{v_1, v_2\}$ is a non-*st* 2-cut and lead to a contradiction.

Finally we show the time complexity. At each iteration, we compute all cut vertices of $G[V - U]$ and check if they have any neighbor in U. By this way we can find all movable vertices or determine to stop the algorithm in linear time since all cut vertices can be found in linear time by Tarjan's algorithm [7]. Therefore it takes $O(|V||E|)$ time in total since there are $O(|V|)$ iterations. $\square$

We conclude the above result by the following theorem.

Theorem 12: If $G = (V, E)$ is biconnected and has no non-*st* 2-cut, there exists a non-separating *st*-path on G which can be found in $O(|V||E|)$ time.

Definition 1: A non-*st* 2-cut $X = \{x_1, x_2\}$ is an HP 2-cut if there are exactly two components in $G - X$ and a Hamiltonian x_1x_2-path on $C + X$ can be found in polynomial time, in which C is one of the components containing neither s nor t.

Corollary 13: If G is biconnected and every non-*st* 2-cut is an HP 2-cut, there exists a non-separating *st*-path which can be found in polynomial time.

Proof: Suppose that $\{s,t\}$ is a 2-cut. By the assumption, there are exactly two components C_1 and C_2 and we can find a Hamiltonian *st*-path on one of them in polynomial time. Apparently the Hamiltonian path is a non-separating *st*-path. Otherwise we assume that $\{s,t\}$ is not a 2-cut.

Let H, T, B_i, D_i and r are the same as defined in the proof of Lemma 11. Again we can show that T is a path and that $r \leq 2$. For each case of r, there is a 2-cut. By the assumption, the corresponding block has a Hamiltonian path which can be found in polynomial time. Thus, each vertex of T either is itself a vertex or can be replaced by a path visiting all vertices of the block exactly once. Concatenating these vertices or paths along T, we can obtain a non-separating st-path.

□

Acknowledgment

This work was supported in part by NSC 97-2221-E-194-064-MY3 and NSC 98-2221-E-194-027-MY3 from the National Science Council, Taiwan.

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